Economic Growth and Unemployment Dynamics

A Comparative Analysis of Okun's Law in Morocco, Brazil, and Spain Using World Bank Data

ALAMI Imrane

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Résumé

This article analyzes the relationship between economic growth and unemployment through a comparative study of Okun's law applied to Morocco, Brazil, and Spain. Using annual data from the World Bank's *World Development Indicators* over the period 1991–2024, the study examines the extent to which real GDP growth explains annual variations in the unemployment rate across different economic contexts. The methodology is based on ordinary least squares (OLS) estimation, complemented by ADF stationarity tests, HAC robust standard errors, and dynamic lagged models. The results show that the coefficient associated with real GDP growth is negative and statistically significant in all three countries, broadly confirming the empirical validity of Okun's law. However, the intensity of this relationship varies substantially : it appears weakest in Morocco, intermediate in Brazil, and strongest in Spain. These findings suggest that economic growth contributes to explaining unemployment fluctuations, but remains insufficient to fully account for the broader dynamics of the labor market.

Keywords : Okun's law, economic growth, unemployment, comparative analysis, Brazil, Morocco, Spain, econometrics, World Bank.

JEL Classification : E24, E32, C22, O54.

1 Introduction

The relationship between economic growth and unemployment constitutes one of the central questions of modern macroeconomics. Since the seminal work of Okun (1962), it has commonly been accepted that an increase in gross domestic product growth should be associated with a decline in the unemployment rate. In other words, when economic activity accelerates, firms produce more, which promotes job creation and reduces unemployment. This relationship, known as Okun's law, has since generated a substantial empirical literature and has been tested across numerous national and regional contexts.

Ball, Leigh, and Loungani (2017) show that Okun's law remains empirically valid in most advanced economies, although the strength of the relationship varies significantly across countries and time periods. Perman, Stephan, and Tavéra (2015), in a meta-analysis covering several dozen international studies, confirm this heterogeneity and emphasize that the Okun coefficient depends strongly on the structure of the labor market, the degree of informality, economic institutions, and the public policies in place. These studies therefore show that Okun's law should not be understood as a perfectly stable universal relationship, but rather as a general tendency whose intensity varies according to context.

However, this relationship should not be considered automatic or mechanical. Economic growth alone does not always explain the evolution of unemployment. Indeed, unemployment also depends on other macroeconomic and structural factors, such as inflation, the real interest rate, the exchange rate, investment, foreign trade, public consumption, population growth, labor productivity, the level of informality, and the institutional functioning of the labor market. Thus, a country may experience positive economic growth without recording a significant decline in unemployment. This phenomenon may be explained by growth that is not sufficiently employment-intensive, by a mismatch between available skills and the actual needs of the labor market, or by institutional rigidities that limit the creation of formal jobs.

In this paper, we propose to examine this relationship through a comparison between Morocco, Brazil, and Spain. These three countries were selected because of the diversity of their economic structures and labor markets. Brazil represents a large emerging economy in Latin America, characterized by significant economic cycles, high income inequality, and a labor market marked by a duality between the formal and informal sectors. Morocco is a North African economy integrated into the MENA region, whose growth depends in particular on agriculture, services, tourism, manufacturing industry, public investment, and phosphate exports. Spain, finally, represents a developed economy within the euro area, whose labor market has historically been sensitive to economic cycles and marked by a strong duality between permanent and temporary contracts.

The research question of this paper can be formulated as follows :

> Is economic growth sufficient to explain variations in unemployment in Morocco, Brazil, and Spain, or is it necessary to take into account other macroeconomic and structural factors ?

Based on this research question, three main hypotheses are formulated :

— **Hypothesis 1 :** Economic growth has a negative effect on the variation in unemployment in the three countries studied, in accordance with Okun's law.

— **Hypothesis 2 :** The strength of the relationship between economic growth and unemployment varies across countries, depending on productive structures, labor market institutions, and the degree of informality.

— **Hypothesis 3 :** Economic growth is not sufficient to fully explain variations in unemployment, which implies the need to consider additional macroeconomic and structural factors.

To address this research question, the study relies on annual data from the World Bank and adopts a rigorous econometric approach. First, Okun's law is estimated using a simple model in which the dependent variable is the annual change in the unemployment rate and the main explanatory variable is annual real GDP growth. The statistical properties of the series are verified using ADF stationarity tests (Dickey and Fuller, 1979), and the estimations are corrected using HAC robust standard errors (Newey and West, 1987) in order to account for potential heteroskedasticity and autocorrelation issues. Second, a dynamic model with lagged variables is estimated in order to assess whether the effect of growth on unemployment is immediate or delayed over time.

This comparative approach makes it possible to evaluate the validity of Okun's law across different economic contexts and to show that the effect of growth on unemployment varies across countries, depending on their productive structures, labor markets, and economic policies. The remainder of the article is organized as follows : Section 2 presents the data and the econometric methodology ; Section 3 reports the empirical results for each of the three countries ; Section 4 provides a comparative discussion of the results ; and Section 5 concludes and proposes avenues for future research.

1.1 Revue de littérature

1.1.1 The Theoretical Foundations of Okun's Law

The relationship between economic growth and unemployment was first formalized by Arthur Okun in 1962. In his seminal work, Okun (1962) empirically established that an increase in real GDP growth is associated with a decline in the unemployment rate, and proposed an empirical rule according to which growth of around 3% is necessary to keep unemployment stable in the United States. This relationship, initially formulated in the American post-war context, has since been generalized and tested across numerous countries and different time periods. It is now considered one of the most widely used analytical frameworks in applied macroeconomics for studying the relationship between economic activity and the labor market.

Two main formulations of Okun's law are commonly used in the empirical literature. The first, known as the differences formulation, relates the annual change in the unemployment rate to annual real GDP growth. The second, known as the gap formulation, relates the unemployment gap from its natural level to the output gap from its potential level. In this study, we retain the differences formulation, which is the most directly estimable using annual data and the most frequently used in international comparative studies.

1.1.2 The International Validity of Okun's Law

Since the work of Okun (1962), numerous studies have sought to test the empirical validity of this relationship in various national contexts. Ball, Leigh, and Loungani (2017) conduct a systematic analysis of Okun's law across a broad sample of advanced economies and show that the relationship remains empirically valid in the vast majority of cases, although the magnitude of the coefficient varies significantly across countries and periods. Their analysis notably highlights that the Okun coefficient tends to be higher in countries where the labor market is more flexible and more sensitive to economic cycles.

Perman, Stephan, and Tavéra (2015) conduct a meta-analysis of several dozen empirical studies devoted to Okun's law in different countries. Their results confirm the heterogeneity of the Okun coefficient at the international level and show that this heterogeneity is largely explained by structural differences between national labor markets, particularly in terms of flexibility, labor market institutions, the degree of informality, and public employment policies. These studies also emphasize that the Okun coefficient tends to vary over time within the same country, suggesting that the relationship between growth and unemployment is not structurally stable.

1.1.3 The Specific Features of Emerging Labor Markets

In emerging and developing economies, the relationship between economic growth and unemployment presents specific features that distinguish it from the case of advanced economies. The significant weight of the informal sector, the predominance of agricultural employment, and the institutional rigidities of the formal labor market often limit the transmission of economic growth into formal employment and mechanically reduce the magnitude of the Okun coefficient measured using official unemployment statistics.

In the Moroccan case, Achy (2010) shows that economic growth has not always been sufficiently intensive in formal employment to significantly reduce unemployment, particularly because of the importance of agriculture, informality, and the mismatch between the education and training system and the actual needs of the labor market. *The Oxford Handbook of the Moroccan Economy* (El Aynaoui et al.) also emphasizes that the structural transformations of the Moroccan economy, although significant, have not yet fully resolved the problem of creating formal and productive jobs for a growing labor force. These studies help explain why the Okun coefficient estimated for Morocco remains relatively weak compared with that observed in more developed economies.

1.1.4 The Specific Features of European Labor Markets

In the European context, and particularly in the Spanish case, the literature highlights structural characteristics that amplify the sensitivity of unemployment to economic cycles. Blanchard and Summers (1986) develop the concept of unemployment hysteresis, according to which negative economic shocks may have lasting and persistent effects on the unemployment rate, even after the return of economic growth. This phenomenon, which is particularly visible in Europe, can be explained by mechanisms such as the loss of human capital among the long-term unemployed, discouragement effects, and insider behavior in the labor market.

Bentolila and Bertola (1990) complement this analysis by showing that institutional rigidities in the labor market, particularly high dismissal costs and the duality between permanent and temporary contracts, amplify the sensitivity of unemployment to economic cycles in Southern European countries. In the Spanish case, these rigidities lead firms to adjust their level of employment mainly by varying the number of temporary contracts, resulting in significant fluctuations in unemployment during periods of economic expansion and recession.

1.1.5 The Econometric Foundations

From a methodological perspective, this study relies on the work of Dickey and Fuller (1979) to verify the stationarity of time series using the ADF test, and on the work of Newey and West (1987) to correct standard errors using the HAC method. These two methodological contributions are essential to ensure the reliability of econometric estimates in the context of macroeconomic time-series analysis. Finally, the probabilistic approach developed by Haavelmo (1944) recalls that any econometric model constitutes a partial and formalized representation of economic reality, and that the results obtained should be interpreted as a statistical measure of an association between variables, rather than as proof of direct causality.

2 Data and Econometric Methodology

2.1 Variables Used

The variables selected in this study correspond to the main variables required for the estimation of Okun's law. The first variable is annual real GDP growth, used as the main explanatory variable. The second variable is the total unemployment rate. Based on this unemployment rate, a third variable is constructed : the annual change in unemployment, denoted `delta_unemployment`. This variable constitutes the dependent variable of the econometric model, since the differences formulation of Okun's law seeks to explain the annual evolution of unemployment rather than its raw level.

The data used in this study come from the World Bank's *World Development Indicators* database. The selected period is 1991–2024, in order to obtain comparable coverage for Morocco, Brazil, and Spain.

TABLE 1 – Variables Used in the Study

Variable	**World Bank Code**	**Description**
`gdp_growth`	`NY.GDP.MKTP.KD.ZG`	Annual real GDP growth, expressed as a percentage.
`unemployment`	`SL.UEM.TOTL.ZS`	Total unemployment rate, expressed as a percentage of the labor force.
`delta_unemployment`	Constructed	Annual change in the unemployment rate, calculated as the difference between unemployment in the current year and unemployment in the previous year.

2.2 Data Availability and Quality

Before proceeding with the econometric estimation, it is necessary to verify data availability. This step makes it possible to identify the number of observations available for each variable, the missing values, as well as the period actually covered by the series. This verification is important because econometric models can only be estimated using complete observations.

2.3 Data Availability for Brazil

Brazil constitutes the first case studied in this analysis. The data used concern annual real GDP growth and the total unemployment rate. Based on the unemployment rate, an additional variable is constructed : the annual change in unemployment, denoted `delta_unemployment`. This variable makes it possible to estimate Okun's law in its standard form.

Table 2 – Data Availability for Brazil

Variable	Observations	Missing Values	First Year	Last Year
gdp_growth	34	0	1991	2024
unemployment	34	0	1991	2024
delta_unemployment	33	1	1992	2024

The table shows that the main variables of the model are available for Brazil over the period 1991–2024. Real GDP growth and the unemployment rate each have 34 observations. The variable `delta_unemployment` contains one fewer observation, since it is calculated as the difference between the unemployment rate of a given year and that of the previous year. The year 1991 is therefore lost during this transformation. The Okun model for Brazil is thus estimated using 33 observations covering the period 1992–2024.

2.4 Stationarity Test for Brazil

An Augmented Dickey-Fuller test is applied to the Brazilian series in order to verify their stationarity before the econometric estimation. This step is important because the use of non-stationary series may lead to misleading econometric results.

Table 3 – ADF Test Results for Brazil

Variable	ADF P-value	Conclusion
gdp_growth	0.0000	Stationary
unemployment	0.0002	Stationary
delta_unemployment	0.0032	Stationary

The results show that real GDP growth, the unemployment rate, and the variable `delta_unemployment` are stationary at the 5% level. The p-value associated with `delta_unemployment` is equal to 0.0032, which confirms the statistical stability of this series. This result justifies its use as the dependent variable in the Okun's law model applied to Brazil.

2.5 Data Availability for Morocco

After the analysis of the Brazilian case, the same approach is applied to Morocco in order to compare the relationship between economic growth and unemployment in a North African context. For Morocco, the available data concern annual real GDP growth and the total unemployment rate over the period 1991–2024. The annual change in unemployment is then constructed from the unemployment rate.

The table shows that the two main variables, namely real GDP growth and the unemployment rate, are available for Morocco over the period 1991–2024. The variable

Table 4 – Data Availability for Morocco

Variable	Available Observations	Missing Values	First Year	Last Year
gdp_growth	34	0	1991	2024
unemployment	34	0	1991	2024
delta_unemployment	33	1	1992	2024

delta_unemployment, which corresponds to the annual change in unemployment, contains one fewer observation because it is calculated as the difference between unemployment in the current year and unemployment in the previous year. Thus, the year 1991 is automatically lost during this transformation.This step helps avoid the use of non-stationary series in the econometric estimation.

2.6 Stationarity Test for Morocco

As in the case of Brazil, an ADF test is applied to the Moroccan series in order to verify their stationarity. This step helps avoid the use of unstable series in the econometric estimation.

Table 5 – ADF Test Results for Morocco

Variable	ADF P-value	Conclusion
gdp_growth	0.0000	Stationary
unemployment	0.6461	Non-stationary
delta_unemployment	0.0000	Stationary

The results show that Morocco's real GDP growth is stationary at the 5% level. In contrast, the raw unemployment rate appears to be non-stationary, with a p-value of 0.6461. This means that the level of unemployment exhibits a persistent dynamic over time. However, the annual change in unemployment is stationary, which justifies its use as the dependent variable in the Okun's law model.

This result confirms the relevance of transforming the unemployment rate into an annual change. The analysis therefore does not seek only to explain the level of unemployment, but rather its evolution from one year to the next.

2.7 Data Availability for Spain

Spain is included in the analysis in order to add a European point of comparison. This case makes it possible to compare the relationship between growth and unemployment in a developed economy whose labor market has historically been sensitive to economic cycles.

Table 6 – Data Availability for Spain

Variable	Available Observations	Missing Values	First Year	Last Year
gdp_growth	34	0	1991	2024
unemployment	34	0	1991	2024
delta_unemployment	33	1	1992	2024

The Spanish data also cover the period 1991–2024 for real GDP growth and the unemployment rate. As in the Moroccan case, the construction of the annual change in unemployment leads to the loss of the first observation. The Okun model for Spain is therefore estimated using 33 available observations over the period 1992–2024.

2.8 Stationarity Test for Spain

The ADF test is also applied to the Spanish series in order to assess their statistical stability before estimating the model.

Table 7 – ADF Test Results for Spain

Variable	ADF P-value	Conclusion
gdp_growth	0.0001	Stationary
unemployment	0.0714	Non-stationary
delta_unemployment	0.0581	Non-stationary

The results indicate that Spanish real GDP growth is stationary at the 5% level. In contrast, the raw unemployment rate is not stationary at this level, with a p-value of 0.0714. The annual change in unemployment has a p-value of 0.0581, which is slightly above the strict 5% threshold but close to the 10% threshold.

This result implies that the estimates for Spain should be interpreted with greater caution than those for Morocco and Brazil. Although the variable `delta_unemployment` is not stationary at the strict 5% level, its proximity to this threshold suggests that the Okun relationship can be analyzed on an indicative basis, while acknowledging the more persistent dynamics of Spanish unemployment. This limitation will be taken into account in the interpretation of the empirical results and in the comparative discussion.

2.9 Comparative Summary of Data Quality

The comparison of the three countries shows that the data required to estimate Okun's law are generally available for Brazil, Morocco, and Spain. However, the results of the stationarity tests reveal important differences across countries.

In Brazil, the core variables of the model appear generally suitable for estimation. In Morocco, the raw unemployment rate is non-stationary, but its annual change becomes stationary, which makes it possible to continue the analysis using the Okun model. In Spain, the situation is more delicate, since the annual change in unemployment remains slightly non-stationary at the 5% level. This means that the Spanish results should be interpreted with greater caution.

These differences confirm that the relationship between economic growth and unemployment cannot be analyzed in a uniform way. It depends not only on the economic dynamics of each country, but also on the statistical properties of the series used.

2.10 Econometric Methodology

This section presents the econometric approach adopted to test the relationship between economic growth and unemployment. The objective is to assess whether real GDP growth can explain variations in unemployment in the countries studied, namely Morocco, Brazil, and Spain.

The analysis first relies on a simple estimation of Okun's law, in which the annual change in unemployment is explained by annual real GDP growth. Then, a dynamic model with lagged variables is estimated in order to verify whether the effect of economic growth on unemployment is immediate or delayed over time.

2.11 Basic Okun's Law Model

The basic model used in this study is based on the following formulation :

$$\Delta u_t = \alpha + \beta g_t + \varepsilon_t \tag{1}$$

where Δu_t represents the annual change in the unemployment rate, g_t represents annual real GDP growth, α is the constant term of the model, β is the Okun coefficient, and ε_t represents the error term.

According to Okun's theory, the coefficient β is expected to be negative. This means that an increase in economic growth should be associated with a decrease in the change in unemployment.

2.12 Estimation Method

The model is estimated using the ordinary least squares method. This method makes it possible to estimate the linear relationship between real GDP growth and the change in unemployment by minimizing the sum of squared errors.

The objective of the estimation is to determine the sign, magnitude, and statistical significance of the coefficient β. If this coefficient is negative and statistically significant, this indicates that the observed relationship is consistent with Okun's law.

In order to strengthen the reliability of the results, the estimations are also corrected using HAC robust standard errors. This correction makes it possible to account for potential heteroskedasticity and autocorrelation problems in the error terms, which may arise in macroeconomic time series. It does not modify the estimated coefficients, but adjusts the standard errors, test statistics, and p-values.

Finally, a dynamic model with lagged variables is estimated in order to verify whether the effect of economic growth on unemployment is immediate or delayed over time. This model introduces real GDP growth lagged by one period as well as the lagged change in unemployment. It therefore makes it possible to assess whether past changes in growth or unemployment contribute to explaining the current change in unemployment.

2.13 Expected Interpretation of the Coefficients

Within the framework of Okun's law, the main coefficient of interest is the coefficient β associated with real GDP growth. A negative coefficient means that an increase in economic growth is associated with a decrease in the annual change in unemployment. Conversely, a positive coefficient would contradict the expected theoretical relationship.

The magnitude of the coefficient also makes it possible to compare the strength of the relationship across countries. For example, a coefficient of -0.10 means that a one-percentage-point increase in real GDP growth is associated, on average, with a 0.10-percentage-point decrease in the annual change in unemployment. The higher the coefficient is in absolute value, the more strongly unemployment reacts to changes in economic growth.

Finally, the statistical significance of the coefficients is evaluated using p-values. A coefficient is considered statistically significant at the 5% level when its p-value is below 0.05. In this case, the estimated relationship can be considered sufficiently robust from a statistical perspective.

2.14 Model Diagnostic Indicators

Several statistical indicators are used to evaluate the quality of the estimated models. The coefficient of determination R^2 measures the share of the variation in the dependent variable explained by the model. A higher R^2 indicates a better explanatory capacity, although it does not, by itself, guarantee the economic validity of the model.

The Durbin-Watson statistic is used to detect potential autocorrelation in the residuals. A value close to 2 generally indicates the absence of significant autocorrelation, whereas a value far from 2 may indicate temporal dependence in the error terms.

The Jarque-Bera test is used to assess the normality of the residuals. A low p-value indicates that the residuals do not follow a normal distribution, which calls for a cautious interpretation of the results. Finally, the AIC and BIC criteria make it possible to compare the relative quality of the models, taking into account both statistical fit and the number of estimated parameters.

3 Empirical Results

3.1 Results for Morocco

This section presents the results of the estimation of Okun's law for Morocco. The dependent variable is the annual change in the unemployment rate, denoted Δu_t, while the main explanatory variable is annual real GDP growth. The objective is to assess whether economic growth helps explain variations in unemployment in the Moroccan case.

TABLE 8 – Model Fit and Diagnostic Indicators of the Basic OLS Model for Morocco

Indicator	**Value**
Number of observations	33
R-squared	0.401
Adjusted R-squared	0.381
F-statistic	20.71
Prob. F-statistic	0.0000774
Log-likelihood	-23.517
AIC	51.03
BIC	54.03
Durbin-Watson	2.365
Jarque-Bera	0.651
Prob. Jarque-Bera	0.722
Condition Number	7.18

3.1.1 Time Evolution of the Main Variables

Before estimating the Okun model, it is necessary to examine the time evolution of the two main variables retained for the Moroccan case : annual real GDP growth and the annual change in the unemployment rate. This preliminary graphical analysis makes it possible to identify the main trends, periods of economic shock, as well as potential responses of unemployment to changes in economic activity.

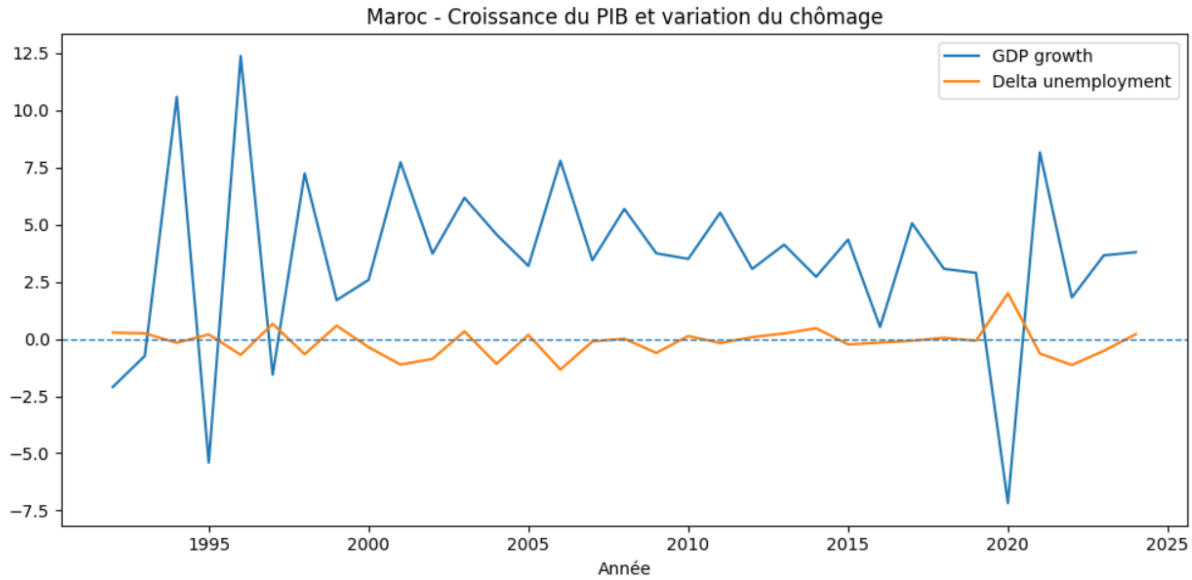


FIGURE 1 – Morocco : GDP Growth and Change in Unemployment

The figure shows that real GDP growth in Morocco exhibits significant fluctuations over the period studied. Some years are marked by strong growth, while others record substantial slowdowns, or even negative growth. These variations reflect the sensitivity of the Moroccan economy to economic cycles, external shocks, and sector-specific characteristics, particularly the importance of agriculture and activities dependent on the economic environment.

The annual change in unemployment, for its part, generally evolves around zero. This means that the unemployment rate does not fluctuate as strongly as GDP growth. However, some years show more visible increases or decreases, indicating that the labor market partially reacts to fluctuations in economic activity.

This preliminary observation suggests that economic growth may play a role in the evolution of unemployment in Morocco, but that this relationship is not perfectly mechanical. In other words, an increase in growth does not always immediately translate into a decline in unemployment. This initial analysis therefore justifies the use of econometric estimation in order to measure more precisely the relationship between GDP growth and changes in unemployment.

3.1.2 Basic Okun's Law Model

The basic model estimated for Morocco is as follows :

$$\Delta u_t = \alpha + \beta g_t + \varepsilon_t \tag{2}$$

where Δu_t represents the annual change in the unemployment rate and g_t represents annual real GDP growth.

Table 9 – Results of the Basic OLS Model for Morocco

Variable	**Coefficient**	**Standard Error**	**t-statistic**	**p-value**
Constant	0.2276	0.119	1.914	0.065
`gdp_growth`	-0.1029	0.023	-4.551	0.000

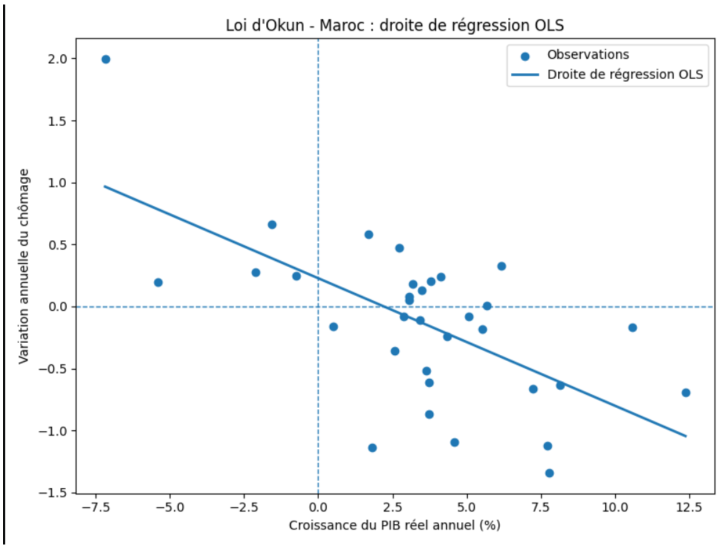


Figure 2 – OLS Regression Line for Okun's Law in Morocco

3.1.3 Assessment of the Predictive Quality of the OLS Model

The following figure compares the actual values of the change in unemployment with the values predicted by the OLS model. The diagonal line represents the perfect prediction line, that is, the case in which the predicted values would be exactly equal to the observed values.

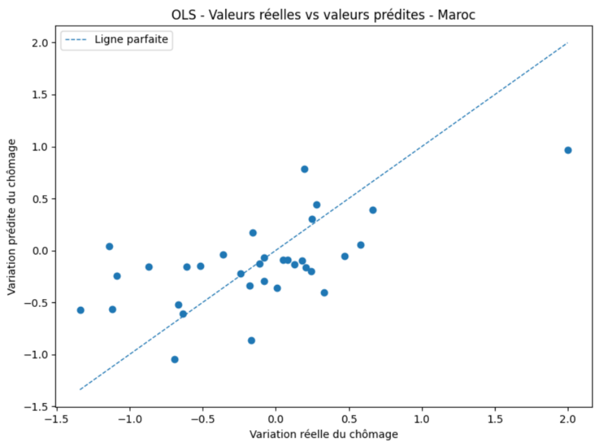
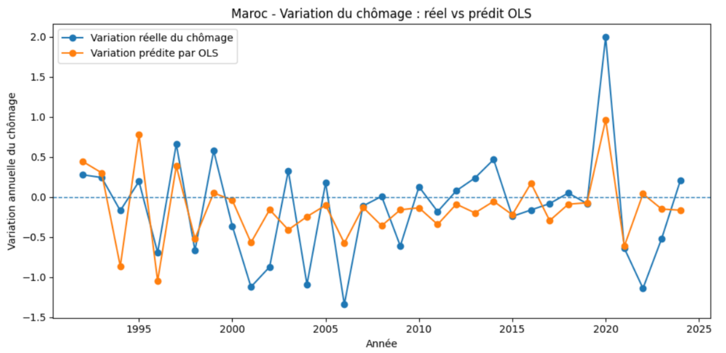


Figure 3 – Actual versus Predicted Values of the Annual Change in Unemployment in Morocco

The graph shows that several observations are relatively close to the perfect prediction line, indicating a partial explanatory capacity of the model. However, the dispersion of the points around this line confirms that GDP growth alone is not sufficient to fully explain variations in unemployment in Morocco.

3.1.4 Comparison Between Actual and Predicted Values

After estimating the basic OLS model, it is useful to compare the actually observed values of the change in unemployment with the values predicted by the model. This comparison makes it possible to visually assess the model's ability to reproduce the evolution of unemployment in Morocco.



The figure shows that the OLS model partially follows the evolution of the change in unemployment, but does not perfectly reproduce all the observed fluctuations. Some years show significant gaps between the actual and predicted values, indicating that GDP growth explains part of the variations in unemployment, but is not sufficient on its own to explain the full dynamics of the Moroccan labor market.

This result remains consistent with the model's R-squared, which indicates a partial explanatory capacity. It therefore confirms the central idea of this study : economic growth plays a role in the evolution of unemployment, but other macroeconomic and structural factors must also be taken into account.

3.1.5 Residual Diagnostics of the OLS Model

The following figure presents the evolution of the residuals of the basic OLS model. The residuals correspond to the difference between the observed values of the change in unemployment and the values predicted by the model.

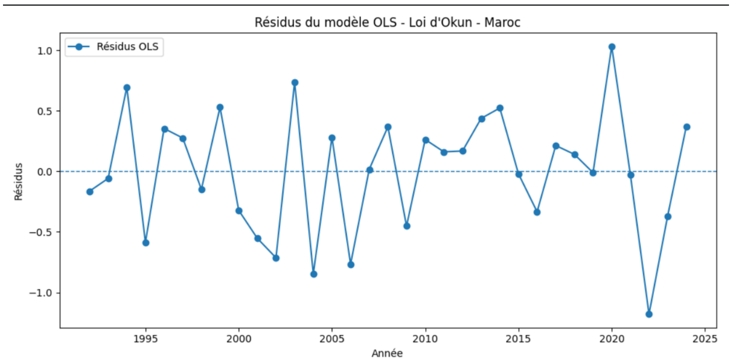


FIGURE 4 – Residuals of the Basic OLS Model for Morocco

The residuals fluctuate around zero, which is generally desirable in a regression model. However, some observations display more pronounced deviations, which justifies the use of a HAC robust correction in order to strengthen the reliability of the statistical inference.

3.1.6 HAC Robust Correction

In order to account for potential heteroskedasticity or autocorrelation in the error terms, a HAC robust correction is also applied to the basic model.

TABLE 10 – Basic Model Results with HAC Robust Standard Errors for Morocco

Variable	**Coefficient**	**HAC Standard Error**	**t-statistic**	**p-value**
Constant	0.2276	0.147	1.553	0.131
`gdp_growth`	-0.1029	0.031	-3.279	0.003

After applying the HAC robust correction, the coefficient of GDP growth remains negative and statistically significant at the 5% level. The p-value associated with the variable `gdp_growth` is equal to 0.003. This confirms the robustness of the main result : in Morocco, an increase in economic growth is associated with a decrease in the change in unemployment.

3.1.7 Dynamic Model with Lagged Variables

In order to verify whether the effect of economic growth on unemployment is immediate or delayed over time, a dynamic model is estimated. This model introduces GDP growth lagged by one period, as well as the lagged change in unemployment :

$$\Delta u_t = \alpha + \beta_0 g_t + \beta_1 g_{t-1} + \rho \Delta u_{t-1} + \varepsilon_t \tag{3}$$

The dynamic model slightly improves the explanatory quality of the model, with the R-squared increasing from 0.401 to 0.476. Current GDP growth remains negative and

Table 11 – Dynamic OLS Model Results for Morocco

Variable	**Coefficient**	**Standard Error**	**t-statistic**	**p-value**
Constant	0.5728	0.206	2.786	0.009
`gdp_growth`	-0.1367	0.028	-4.854	0.000
`gdp_growth_lag1`	-0.0711	0.034	-2.081	0.047
`delta_unemployment_lag1`	-0.2319	0.180	-1.291	0.207

Table 12 – Model Fit and Diagnostic Indicators of the Dynamic Model for Morocco

Indicator	**Value**
Number of observations	32
R-squared	0.476
Adjusted R-squared	0.420
F-statistic	8.475
Prob. F-statistic	0.000365
Durbin-Watson	2.094
Jarque-Bera	1.430
Prob. Jarque-Bera	0.489
Condition Number	14.8

significant, with a coefficient of -0.1367. GDP growth lagged by one period is also negative and significant at the 5% level, with a p-value of 0.047.

These results suggest that the effect of economic growth on unemployment in Morocco is not only immediate. Part of this effect may also appear with a delay. In contrast, the lagged change in unemployment is not statistically significant, indicating that the past dynamics of unemployment do not strongly explain its current variation in this specification.

3.1.8 Summary of the Results for Morocco

The results obtained for Morocco broadly confirm the empirical validity of Okun's law. In the basic model, the coefficient of GDP growth is negative and significant, indicating that an increase in economic growth is associated with a decrease in the change in unemployment.

The HAC robust correction confirms this result, since the coefficient remains significant despite the adjustment of the standard errors. The dynamic model also shows that both current growth and lagged growth have a negative effect on the change in unemployment. This suggests that economic growth plays an important role in explaining unemployment in Morocco, but that its effect may appear both immediately and with a certain time lag.

However, the model's R-squared remains below 50%, which means that economic growth alone is not sufficient to fully explain variations in unemployment. Other structural factors, such as job quality, informality, the match between education and the labor market, public policies, and the sectoral structure of the economy, must also be taken into account.

A one-percentage-point increase in real GDP growth is associated with an approximately 0.10-percentage-point decrease in the annual change in unemployment in the basic model.

3.2 Results for Brazil

This section presents the results of the estimation of Okun's law for Brazil. As in the case of Morocco, the dependent variable is the annual change in the unemployment rate, denoted Δu_t, while the main explanatory variable is annual real GDP growth. The objective is to assess whether economic growth helps explain variations in unemployment in the Brazilian context.

3.2.1 Time Evolution of the Main Variables

Before estimating the Okun model, it is useful to examine the time evolution of real GDP growth and the annual change in unemployment in Brazil.

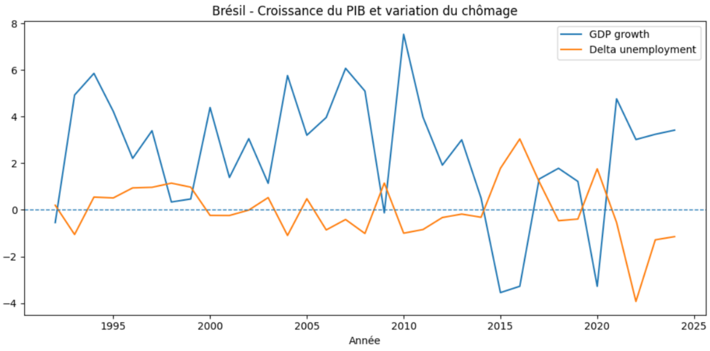


FIGURE 5 – Evolution of Real GDP Growth and the Annual Change in Unemployment in Brazil

The figure shows that Brazilian GDP growth experienced significant fluctuations over the period studied, with some years marked by economic slowdowns or contractions. The change in unemployment also displays substantial movements, particularly around periods of economic shock. This initial observation suggests the existence of a relationship between economic growth and the evolution of unemployment, but this relationship must be tested econometrically.

3.2.2 Basic Okun's Law Model

The basic model estimated for Brazil is as follows :

$$\Delta u_t = \alpha + \beta g_t + \varepsilon_t \tag{4}$$

where Δu_t represents the annual change in unemployment and g_t represents annual real GDP growth.

TABLE 13 – Basic OLS Model Results for Brazil

Variable	**Coefficient**	**Standard Error**	**t-statistic**	**p-value**
Constant	0.7009	0.231	3.031	0.005
`gdp_growth`	-0.2887	0.064	-4.491	0.000

The results of the basic OLS model show that the coefficient associated with GDP growth is negative and statistically significant. The estimated coefficient is equal to -0.2887, which means that a one-percentage-point increase in real GDP growth is associated, on average, with a 0.2887-percentage-point decrease in the annual change in unemployment.

This result is consistent with Okun's law. The p-value associated with the variable `gdp_growth` is below the 5% threshold, indicating that the relationship is statistically significant. The model's R-squared is equal to 0.394, which means that approximately

Table 14 – Model Fit and Diagnostic Indicators of the Basic OLS Model for Brazil

Indicator	**Value**
Number of observations	33
R-squared	0.394
Adjusted R-squared	0.375
F-statistic	20.17
Prob. F-statistic	0.0000918
Log-likelihood	-45.039
AIC	94.08
BIC	97.07
Durbin-Watson	1.365
Jarque-Bera	50.974
Prob. Jarque-Bera	0.00000000000853
Condition Number	5.07

39.4% of the annual variations in unemployment in Brazil are explained by real GDP growth in this simple specification.

However, it should be noted that the Jarque-Bera test indicates non-normality of the residuals in the Brazilian model. The Jarque-Bera statistic is high and its p-value is close to zero, suggesting that the residuals do not follow a normal distribution. This situation may be linked to the presence of major macroeconomic shocks over the period studied, particularly the Brazilian economic crisis of 2015–2016 and the COVID-19 pandemic in 2020. This result does not necessarily invalidate the estimated relationship, but it calls for a cautious interpretation of the results. It also justifies the use of HAC robust standard errors in order to strengthen the reliability of the statistical inference.

3.2.3 Visualization of the Okun Relationship

The following figure provides a visual representation of the relationship between real GDP growth and the annual change in unemployment in Brazil. The points represent the annual observations, while the line corresponds to the estimated OLS regression.

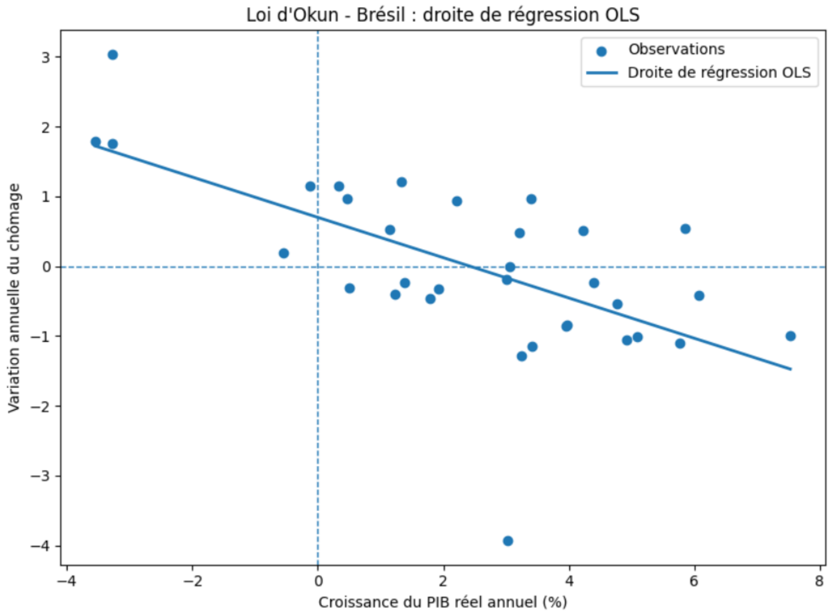


FIGURE 6 – OLS Regression Line for Okun's Law in Brazil

The negative slope of the line visually confirms the inverse relationship between economic growth and the change in unemployment. Years of strong growth are generally associated with a decline in the change in unemployment, which is consistent with Okun's law.

3.2.4 Assessment of the Predictive Quality of the OLS Model

The following figure compares the evolution of the actual change in unemployment with the values predicted by the OLS model. This comparison makes it possible to visually assess the model's ability to reproduce the observed fluctuations in unemployment.

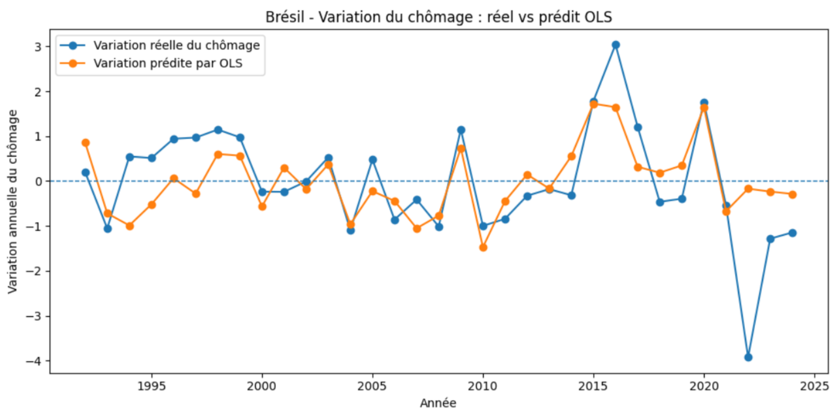


FIGURE 7 – Actual and Predicted Values of the Annual Change in Unemployment in Brazil

The graph shows that the model partially follows the evolution of the change in unemployment, but does not perfectly reproduce all the observed fluctuations. Some years display substantial gaps between the actual and predicted values. This confirms that GDP growth explains an important part of the variations in unemployment, but is not sufficient on its own to explain the full dynamics of the Brazilian labor market.

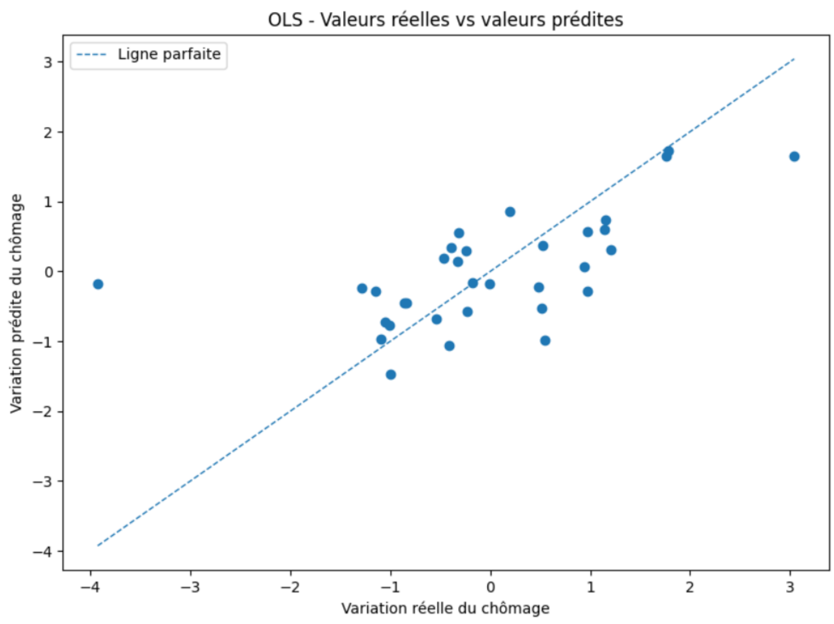


FIGURE 8 – Observed versus Predicted Values of the Annual Change in Unemployment in Brazil

This second graph directly compares the observed values with the predicted values. The closer the points are to the diagonal line, the more accurate the model's prediction. The dispersion of the points confirms that the model has a partial, though imperfect, explanatory capacity.

3.2.5 Residual Diagnostics of the OLS Model

The analysis of the residuals makes it possible to assess the gaps between the observed values and the values predicted by the model. Ideally, the residuals should fluctuate around zero without displaying any systematic structure.

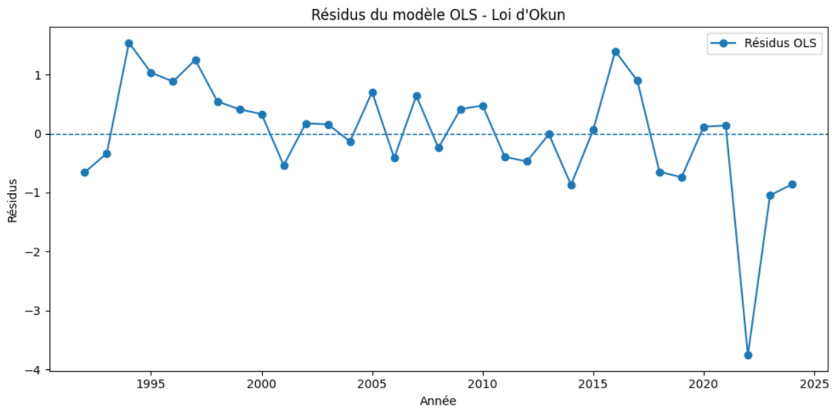


FIGURE 9 – Residuals of the Basic OLS Model for Brazil

The residuals fluctuate overall around zero, but some observations display substantial deviations. This suggests that the simple model does not fully capture the variations in unemployment. This observation justifies the use of a HAC robust correction in order to strengthen the reliability of the statistical inference.

3.2.6 HAC Robust Correction

In order to account for potential heteroskedasticity or autocorrelation in the error terms, a HAC robust correction is applied to the basic model.

TABLE 15 – Basic Model Results with HAC Robust Standard Errors for Brazil

Variable	**Coefficient**	**HAC Standard Error**	**t-statistic**	**p-value**
Constant	0.7009	0.214	3.273	0.003
`gdp_growth`	-0.2887	0.050	-5.728	0.000

After the HAC robust correction, the coefficient of GDP growth remains negative and statistically significant. The p-value associated with `gdp_growth` remains below the 5% threshold, which confirms the robustness of the main result. Thus, in the case of Brazil, the inverse relationship between economic growth and the change in unemployment remains valid after correcting the standard errors.

3.2.7 Dynamic Model with Lagged Variables

In order to verify whether the effect of economic growth on unemployment is immediate or delayed, a dynamic model is estimated. This model introduces GDP growth lagged by one period, as well as the lagged change in unemployment :

$$\Delta u_t = \alpha + \beta_0 g_t + \beta_1 g_{t-1} + \rho \Delta u_{t-1} + \varepsilon_t \tag{5}$$

Table 16 – Dynamic OLS Model Results for Brazil

Variable	Coefficient	Standard Error	t-statistic	p-value
Constant	0.6764	0.286	2.368	0.025
`gdp_growth`	-0.2886	0.066	-4.400	0.000
`gdp_growth_lag1`	0.0144	0.082	0.175	0.863
`delta_unemployment_lag1`	0.3072	0.177	1.734	0.094

Table 17 – Model Fit and Diagnostic Indicators of the Dynamic Model for Brazil

Indicator	Value
Number of observations	32
R-squared	0.484
Adjusted R-squared	0.428
F-statistic	8.747
Prob. F-statistic	0.000298
Log-likelihood	-41.592
AIC	91.18
BIC	97.05
Durbin-Watson	1.866
Jarque-Bera	75.818
Prob. Jarque-Bera	0.0000000000000000344
Condition Number	8.43

The dynamic model improves the explanatory quality of the model, with the R-squared increasing from 0.394 to 0.484. Current GDP growth remains negative and statistically significant, with a coefficient of -0.2886. By contrast, GDP growth lagged by one period is not significant, which suggests that the effect of growth on unemployment in Brazil is mainly immediate in this specification.

The lagged change in unemployment has a p-value of 0.094 in the dynamic OLS model, which makes it non-significant at the 5% level, but close to the 10% threshold. This may indicate some persistence in unemployment dynamics, although the result is not sufficiently strong in the non-robust specification.

3.2.8 HAC Robust Correction of the Dynamic Model

A HAC robust correction is also applied to the dynamic model in order to verify the stability of the results.

Table 18 – Dynamic Model Results with HAC Robust Standard Errors for Brazil

Variable	**Coefficient**	**HAC Standard Error**	**t-statistic**	**p-value**
Constant	0.6764	0.163	4.154	0.000
`gdp_growth`	-0.2886	0.039	-7.440	0.000
`gdp_growth_lag1`	0.0144	0.062	0.232	0.818
`delta_unemployment_lag1`	0.3072	0.083	3.690	0.001

After the HAC correction, current GDP growth remains negative and strongly significant. Lagged growth remains non-significant. By contrast, the lagged change in unemployment becomes significant at the 5% level, which suggests some persistence in unemployment once the standard errors are corrected.

3.2.9 Summary of the Results for Brazil

The results obtained for Brazil broadly confirm the empirical validity of Okun's law. In the basic OLS model, the coefficient associated with GDP growth is negative and statistically significant. This means that an increase in economic growth is associated with a decrease in the annual change in unemployment.

The HAC robust correction confirms this result, since the GDP growth coefficient remains negative and significant. The dynamic model improves the explanatory quality of the model, but shows that the main effect comes primarily from current growth rather than lagged growth.

However, the R-squared of the basic model remains below 50%. This means that economic growth alone is not sufficient to fully explain variations in unemployment in Brazil. Other macroeconomic and structural factors, such as the structure of the labor market, informality, public policies, investment, and productivity, must also be taken into account.

3.3 Results for Spain

This section presents the results of the estimation of Okun's law for Spain. As in the cases of Morocco and Brazil, the dependent variable is the annual change in the unemployment rate, denoted Δu_t, while the main explanatory variable is annual real GDP growth. The objective is to assess whether economic growth helps explain variations in unemployment in the Spanish case.

3.3.1 Time Evolution of the Main Variables

Before estimating the Okun model, it is useful to examine the time evolution of real GDP growth and the annual change in unemployment in Spain.

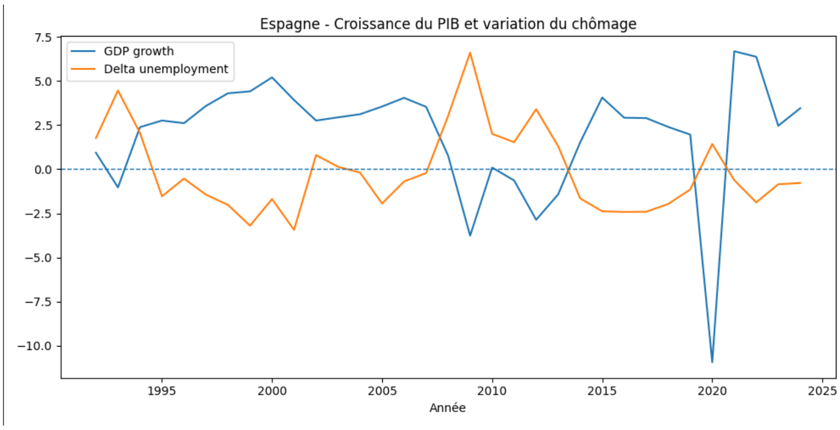


FIGURE 10 – Evolution of Real GDP Growth and the Annual Change in Unemployment in Spain

The figure shows that Spanish GDP growth experienced significant fluctuations over the period studied. The change in unemployment also displays marked movements, suggesting a strong sensitivity of the Spanish labor market to economic cycles. This initial observation justifies the econometric estimation of Okun's law in order to measure this relationship more precisely.

3.3.2 Basic Okun's Law Model

The basic model estimated for Spain is as follows :

$$\Delta u_t = \alpha + \beta g_t + \varepsilon_t \tag{6}$$

where Δu_t represents the annual change in unemployment and g_t represents annual real GDP growth.

Table 19 – Basic OLS Model Results for Spain

Variable	**Coefficient**	**Standard Error**	**t-statistic**	**p-value**
Constant	0.7562	0.363	2.085	0.045
`gdp_growth`	-0.4540	0.095	-4.756	0.000

Table 20 – Model Fit and Diagnostic Indicators of the Basic OLS Model for Spain

Indicator	**Value**
Number of observations	33
R-squared	0.422
Adjusted R-squared	0.403
F-statistic	22.62
Prob. F-statistic	0.0000432
Log-likelihood	-64.857
AIC	133.7
BIC	136.7
Durbin-Watson	1.180
Jarque-Bera	0.140
Prob. Jarque-Bera	0.932
Condition Number	4.53

The results of the basic OLS model show that the coefficient associated with GDP growth is negative and statistically significant. The estimated coefficient is equal to -0.4540, which means that a one-percentage-point increase in real GDP growth is associated, on average, with a 0.4540-percentage-point decrease in the annual change in unemployment.

This result is consistent with Okun's law. The model's R-squared is equal to 0.422, which means that approximately 42.2% of the annual variations in unemployment in Spain are explained by real GDP growth in this simple specification.

3.3.3 Visualization of the Okun Relationship

The following figure provides a visual representation of the relationship between real GDP growth and the annual change in unemployment in Spain. The points represent the annual observations, while the line corresponds to the estimated OLS regression.

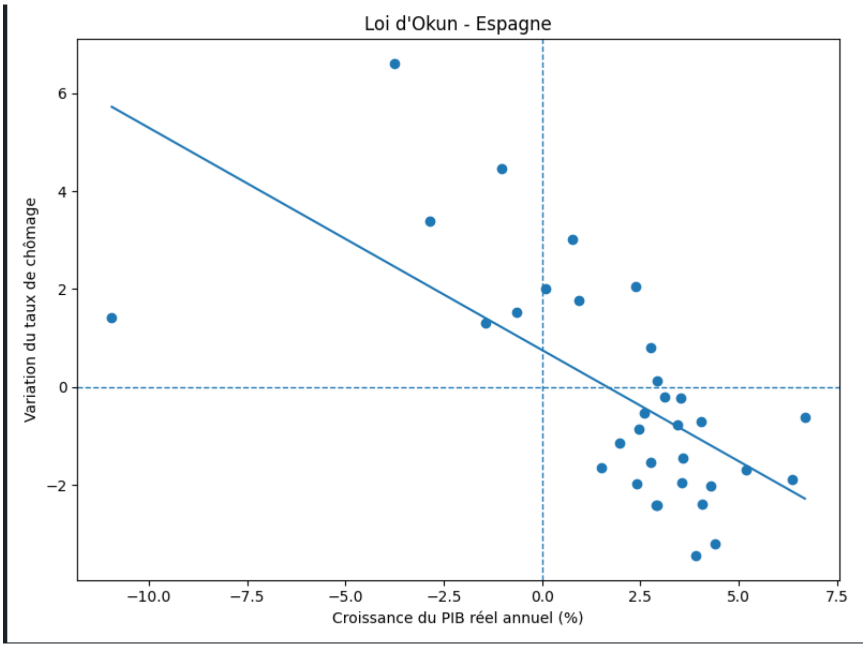


FIGURE 11 – OLS Regression Line for Okun's Law in Spain

The negative slope of the line visually confirms the inverse relationship between economic growth and the change in unemployment. This means that years of higher growth are generally associated with a decline in the change in unemployment.

3.3.4 Assessment of the Predictive Quality of the OLS Model

The following figure compares the evolution of the actual change in unemployment with the values predicted by the OLS model. This comparison makes it possible to visually assess the model's ability to reproduce the observed fluctuations in unemployment.

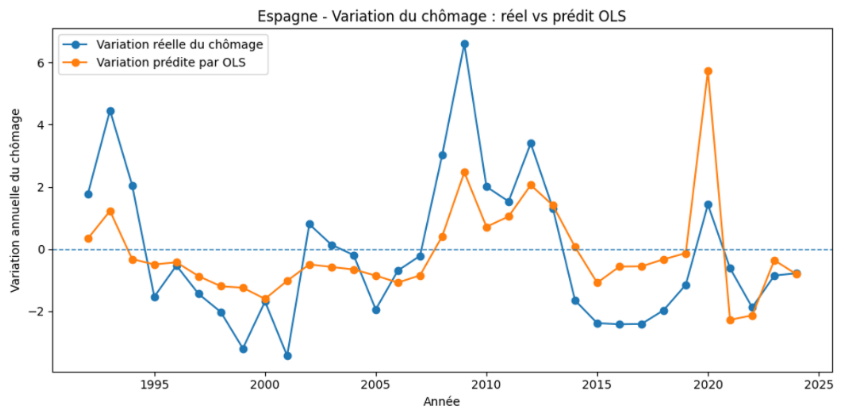


FIGURE 12 – Actual and Predicted Values of the Annual Change in Unemployment in Spain

The graph shows that the model partially follows the evolution of the change in unemployment, but does not perfectly reproduce all the observed fluctuations. This confirms that economic growth explains an important part of the variations in unemployment, without being sufficient on its own to explain the full dynamics of the Spanish labor market.

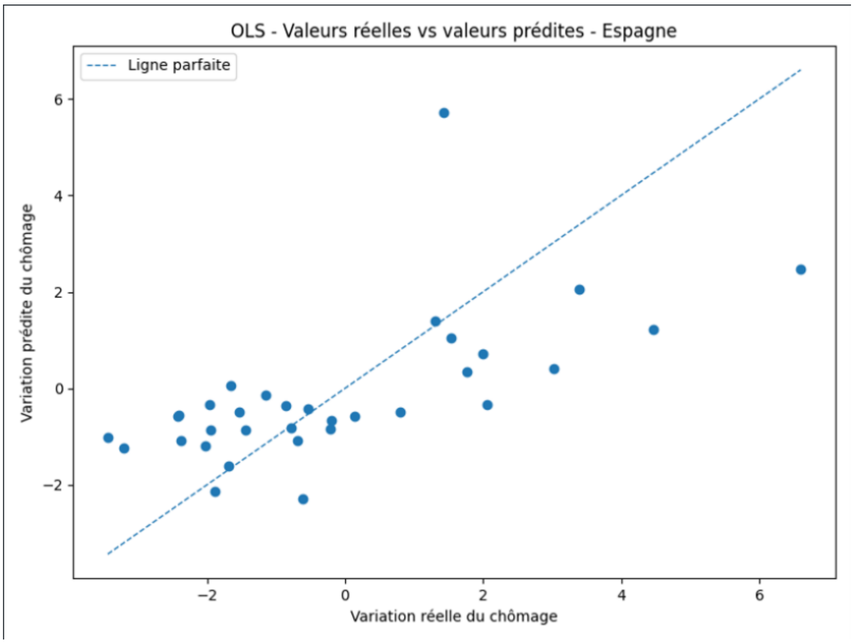


FIGURE 13 – Observed versus Predicted Values of the Annual Change in Unemployment in Spain

3.3.5 Residual Diagnostics of the OLS Model

The analysis of the residuals makes it possible to assess the gaps between the observed values and the values predicted by the model. Ideally, the residuals should fluctuate around zero without displaying any systematic structure.

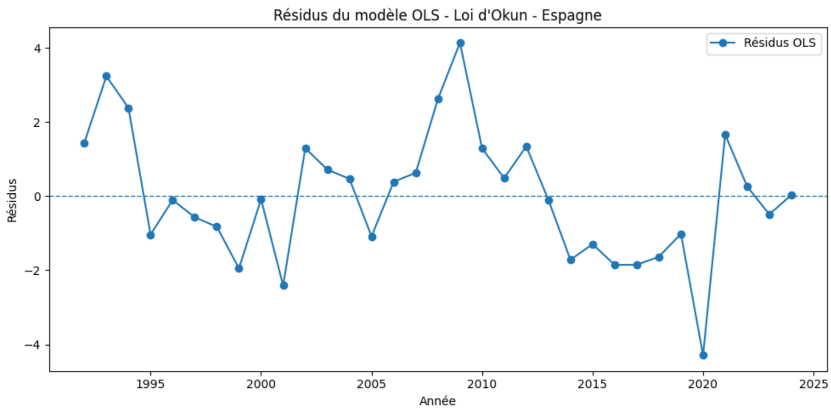


FIGURE 14 – Residuals of the Basic OLS Model for Spain

The residuals fluctuate overall around zero, but some observations display more pronounced deviations. This observation justifies the use of a HAC robust correction in order to verify the statistical robustness of the results.

3.3.6 HAC Robust Correction

In order to account for potential heteroskedasticity or autocorrelation in the error terms, a HAC robust correction is applied to the basic model.

TABLE 21 – Basic Model Results with HAC Robust Standard Errors for Spain

Variable	**Coefficient**	**HAC Standard Error**	**t-statistic**	**p-value**
Constant	0.7562	0.639	1.183	0.246
`gdp_growth`	-0.4540	0.193	-2.351	0.025

After applying the HAC robust correction, the coefficient of GDP growth remains negative and statistically significant at the 5% level. The p-value associated with `gdp_growth` is equal to 0.025. This confirms that the inverse relationship between economic growth and the change in unemployment remains robust in the Spanish case.

3.3.7 Dynamic Model with Lagged Variables

In order to verify whether the effect of economic growth on unemployment is immediate or delayed, a dynamic model is estimated. This model introduces GDP growth lagged by one period, as well as the lagged change in unemployment :

$$\Delta u_t = \alpha + \beta_0 g_t + \beta_1 g_{t-1} + \rho \Delta u_{t-1} + \varepsilon_t \tag{7}$$

TABLE 22 – Dynamic OLS Model Results for Spain

Variable	**Coefficient**	**Standard Error**	**t-statistic**	**p-value**
Constant	0.5056	0.388	1.303	0.203
`gdp_growth`	-0.3497	0.089	-3.946	0.000
`gdp_growth_lag1`	0.0264	0.110	0.239	0.813
`delta_unemployment_lag1`	0.4566	0.164	2.779	0.010

The dynamic model clearly improves the explanatory quality of the model, with the R-squared increasing from 0.422 to 0.592. Current GDP growth remains negative and statistically significant, with a coefficient of -0.3497. By contrast, GDP growth lagged by one period is not significant, which suggests that the effect of growth on unemployment is mainly immediate.

Table 23 – Model Fit and Diagnostic Indicators of the Dynamic Model for Spain

Indicator	**Value**
Number of observations	32
R-squared	0.592
Adjusted R-squared	0.549
F-statistic	13.55
Prob. F-statistic	0.0000120
Log-likelihood	-57.444
AIC	122.9
BIC	128.7
Durbin-Watson	1.851
Jarque-Bera	1.654
Prob. Jarque-Bera	0.437
Condition Number	6.94

The lagged change in unemployment is positive and significant, with a p-value of 0.010. This indicates a certain persistence of unemployment in Spain : past changes in unemployment contribute to explaining current changes.

3.3.8 Summary of the Results for Spain

The results obtained for Spain broadly confirm the empirical validity of Okun's law. In the basic OLS model, the coefficient associated with GDP growth is negative and statistically significant. This means that an increase in economic growth is associated with a decrease in the annual change in unemployment.

The HAC robust correction confirms this result, since the GDP growth coefficient remains negative and significant. The dynamic model improves the explanatory quality of the model and highlights a certain persistence of unemployment, since the lagged change in unemployment is significant.

However, despite these results supporting Okun's law, economic growth alone is not sufficient to fully explain variations in unemployment in Spain. The Spanish labor market presents a specific dynamic, marked by strong sensitivity to economic cycles and by a certain persistence of unemployment.

4 Comparative Discussion

4.1 Comparative Summary of the Empirical Results

The following table summarizes the main results obtained for the three countries studied.

Table 24 – Comparative Summary of Okun's Law Results

Country	Okun Coefficient	R^2 of the Basic Model	Dynamic Result	Interpretation
Morocco	−0.1029	0.401	Current and lagged growth are significant	Negative relationship, but relatively weak
Brazil	−0.2887	0.394	Mainly immediate effect of growth	Negative relationship of intermediate intensity
Spain	−0.4540	0.422	Significant persistence of unemployment	Strong negative relationship and labor market highly sensitive to cycles

This summary shows that Okun's law is empirically verified in the three countries, since the coefficient associated with real GDP growth is always negative. However, the intensity of this relationship varies considerably. Spain displays the strongest reaction of unemployment to growth, followed by Brazil, while Morocco shows a more limited reaction.

This table highlights a clear hierarchy among the three countries studied. Spain has the highest Okun coefficient in absolute value, followed by Brazil and then Morocco. This difference suggests that the transmission of economic growth to the labor market varies significantly depending on the economic and institutional structures specific to each country.

4.2 A Partial Validation of Okun's Law

The empirical results obtained for Morocco, Brazil, and Spain broadly confirm the relationship expected by Okun's law, in line with the conclusions of Ball, Leigh, and

Loungani (2017). In all three countries, the coefficient associated with real GDP growth is negative and statistically significant, indicating that an increase in economic growth is associated with a decrease in the annual change in the unemployment rate. This result is robust across the different specifications tested, particularly after correcting standard errors using the HAC method of Newey and West (1987).

However, the intensity of this relationship varies substantially across countries, which is consistent with the meta-analysis of Perman, Stephan, and Tavéra (2015). The Okun coefficient estimated in the basic model is -0.10 for Morocco, -0.29 for Brazil, and -0.45 for Spain. This hierarchy clearly illustrates that the same increase in economic growth does not produce the same effect on unemployment depending on the country considered. Spain shows the strongest reaction, which can be linked to labor market flexibility, the sensitivity of the construction sector to economic cycles, and the hysteresis effects documented by Blanchard and Summers (1986). Brazil displays an intermediate reaction, influenced by the magnitude of economic cycles and the structure of the Brazilian labor market. Morocco, finally, presents the weakest reaction, reflecting the limited role of formal growth in the creation of statistically measured jobs, particularly because of the weight of informality and agriculture.

These differences confirm that Okun's law is not a perfectly stable universal relationship. It operates empirically in the three cases, but with different levels of intensity depending on each country's productive structures, labor market institutions, and economic policies. Moreover, the R^2 values obtained in the basic models, ranging between 0.39 and 0.42, show that GDP growth explains only part of the variations in unemployment. In all three countries, an important share of unemployment dynamics therefore remains determined by other factors not captured by the simple Okun model.

4.3 Haavelmo's Contribution : The Econometric Model as a Partial Representation of Reality

The results obtained in this study must be interpreted with caution, in light of the probabilistic approach developed by Trygve Haavelmo (1944). In his seminal work, Haavelmo emphasizes that an econometric model does not directly reproduce economic reality in all its complexity. Rather, it constitutes a statistical and probabilistic representation of a relationship between observable variables, constructed on the basis of simplifying assumptions about the structure of the economy. This distinction is fundamental for correctly interpreting the results of an econometric estimation.

In the context of this study, the Okun model makes it possible to measure a statistical relationship between two fundamental macroeconomic variables : annual real GDP growth and the annual change in the unemployment rate. However, this estimated relationship does not mean that growth is the only cause of unemployment, nor that it is its direct and exclusive cause. The error term of the model precisely represents everything that is not explained by GDP growth in the selected specification : external economic shocks, institutional changes, public employment policies, the sectoral structure of the economy, the degree of informality, demographic transformations, productivity variations, and composition effects within the labor force.

This perspective is particularly important in the context of a comparative study involving countries as different as Morocco, Brazil, and Spain. Each of these countries has its own institutional, structural, and economic characteristics, which influence the way in which growth is transmitted to the labor market. From this perspective, the econometric model provides a simplified but useful reading of reality. It makes it possible to identify a statistically significant relationship and to compare its intensity across countries, but it should not be confused with a complete and causal explanation of the economic phenomenon under study.

In other words, the results confirm a relationship compatible with Okun's law in the three countries, but they are not sufficient to establish that economic growth fully explains unemployment. This limitation is inherent to any econometric approach based on a simplified model, and it highlights the importance of combining quantitative analysis with a qualitative and structural reading of the real economy. It is precisely in this spirit that the following sections place the econometric results within their respective national contexts.

4.4 The Moroccan Case in Light of *The Oxford Handbook of the Moroccan Economy*

The Moroccan case particularly illustrates the limits of an explanation of unemployment based solely on economic growth. The Okun coefficient estimated for Morocco is negative and statistically significant, confirming the existence of an inverse relationship between GDP growth and the change in unemployment. However, this coefficient, equal to -0.10 in the basic model, remains substantially lower in absolute value than those estimated for Brazil (-0.29) and Spain (-0.45). This means that, for the same one-percentage-point increase in economic growth, the reduction in unemployment is much more limited in Morocco than in the two other countries.

This relative weakness of the Moroccan Okun coefficient can be linked to several structural characteristics of the Moroccan economy. First, the significant weight of agriculture in GDP and rural employment introduces strong volatility into growth, particularly depending on climatic conditions and annual rainfall. However, favorable agricultural growth does not necessarily translate into a reduction in statistically measured urban unemployment, since it mainly concerns rural workers who are often classified as self-employed workers or unpaid family workers rather than as unemployed in the strict sense. Second, the high level of informality in the Moroccan economy means that a significant share of the employment created during growth phases escapes official unemployment statistics, which mechanically weakens the measured relationship between growth and changes in unemployment.

The Oxford Handbook of the Moroccan Economy (El Aynaoui et al.) highlights the importance of structural transformations in the Moroccan economy, particularly through massive public investment in infrastructure, industrial development, export-oriented sectors such as the automotive and aeronautics industries, and economic diversification policies. These transformations have helped improve the long-term growth potential of the Moroccan economy. However, as Achy (2010) emphasizes, transforming this growth into formal, stable, and well-paid jobs remains a central challenge for the Moroccan economy.

Unemployment may therefore persist despite positive growth when the jobs created are concentrated in sectors with low labor intensity, when informality absorbs a significant share of the labor force without reducing statistically measured unemployment, or when the qualifications of available workers do not correspond to the actual needs of expanding firms. This phenomenon of mismatch between labor supply and labor demand is particularly visible in the case of unemployment among young graduates in Morocco, which remains structurally high despite phases of economic growth. The result of the dynamic model, which shows that the effect of growth on unemployment appears both immediately and with a certain time lag, is also consistent with this reality : adjustments in the Moroccan labor market are slow and partial, reflecting the structural and institutional rigidities of the economy.

Thus, the Moroccan case confirms that economic growth is a necessary but insufficient condition for sustainably reducing unemployment. For growth to effectively translate into a significant and lasting decline in the unemployment rate, it must be inclusive, intensive in formal employment, and accompanied by appropriate public policies, particularly in the areas of vocational training, the development of the formal private sector, and labor market reform.

4.5 Comparison with Brazil and Spain

The comparison of the results obtained for Brazil and Spain with those obtained for Morocco highlights important differences in the way economic growth is transmitted to the labor market across national contexts.

In the case of Brazil, the Okun coefficient estimated in the basic model is -0.29, which is nearly three times higher in absolute value than that of Morocco. This result suggests that the Brazilian labor market reacts more strongly to variations in economic activity. This higher sensitivity can be explained by several factors specific to the Brazilian economy. On the one hand, Brazil experienced particularly pronounced economic cycles over the period studied, with phases of strong growth followed by major contractions, notably during the 2015–2016 economic crisis and the COVID-19 pandemic in 2020. These major macroeconomic shocks led to significant fluctuations in unemployment, which tends to mechanically strengthen the estimated relationship between growth and unemployment. On the other hand, the Brazilian labor market displays a certain degree of flexibility in employment adjustments, particularly through temporary contracts and mechanisms for adjusting working hours, which allows variations in economic activity to be transmitted more rapidly to the labor market.

However, the Brazilian dynamic model shows that the effect of growth on unemployment is mainly immediate, since lagged growth is not statistically significant. By contrast, after applying the HAC robust correction, the lagged change in unemployment becomes significant, suggesting some persistence in unemployment dynamics in Brazil. This result indicates that past variations in unemployment contribute to explaining current variations, which may reflect worker discouragement effects, losses of human capital, or rigidities in certain segments of the Brazilian labor market.

In the case of Spain, the estimated Okun coefficient is -0.45, the highest in absolute value among the three countries studied. This result confirms that the Spanish labor market is particularly sensitive to economic cycles, which is consistent with the extensive literature on the specific features of the Spanish labor market. Blanchard and Summers (1986) highlighted the phenomenon of hysteresis in European labor markets, according to which negative economic shocks may have lasting effects on unemployment, even after the return of economic growth. This phenomenon is particularly visible in Spain, where the unemployment rate reached very high levels during the 2008–2009 financial crisis, with persistent effects on employment for several years after the beginning of the economic recovery.

Bentolila and Bertola (1990) explain this high sensitivity through the institutional rigidities of the Spanish labor market, particularly high dismissal costs for workers under permanent contracts and the strong duality between permanent and temporary contracts. This duality leads firms to adjust their employment levels mainly by varying the number of temporary contracts, which amplifies fluctuations in employment and unemployment in response to economic cycles. The Spanish dynamic model confirms this analysis by highlighting significant unemployment persistence : the lagged change in unemployment is positive and statistically significant, with a coefficient of 0.4566 and a p-value of 0.010. This means that past increases in unemployment tend to persist over time, reflecting hysteresis effects and the structural difficulties faced by the long-term unemployed in reintegrating into the Spanish labor market.

The comparison of the three countries therefore confirms that Okun's law operates empirically in very different contexts, but that its intensity depends strongly on the institutional, productive, and social structures specific to each economy. This heterogeneity recalls that any economic policy aiming to reduce unemployment through growth must take into account the national specificities of the labor market.

4.6 Is Economic Growth Sufficient to Explain Unemployment ?

The empirical results obtained in this study make it possible to provide a nuanced answer to the central research question formulated in the introduction. Economic growth does indeed contribute to explaining variations in unemployment in the three countries studied, as shown by the negative and statistically significant coefficients obtained for Morocco, Brazil, and Spain. In this sense, Okun's law constitutes a relevant and empirically validated analytical framework for understanding the relationship between economic activity and the labor market.

However, the R^2 values obtained in the basic models, ranging between 0.39 and 0.42, clearly indicate that real GDP growth explains only around 40% of the annual variations in unemployment in each of the three countries. In other words, more than half of unemployment dynamics remains determined by factors that are not captured by the simple Okun model. This observation is fundamental : it shows that economic growth is a necessary but insufficient condition for fully understanding and explaining the evolution of unemployment.

In the Moroccan case, the complementary factors that could improve the explanation of unemployment include, in particular, the degree of informality in the economy, the sectoral structure of growth, unemployment among young graduates, the mismatch between the education and training system and the actual needs of the labor market, the low participation of women in the formal labor market, and the quality of jobs created during growth phases. These structural factors, which are often difficult to quantify precisely, play a decisive role in the transmission of growth to formal employment and explain why the Moroccan Okun coefficient remains relatively weak.

In the Brazilian case, income inequalities, economic cycles amplified by external shocks, monetary and fiscal policies, investment, and labor productivity constitute important complementary factors for explaining variations in unemployment beyond GDP growth alone. The persistence of unemployment highlighted by the dynamic model with HAC correction also suggests that mechanisms of temporal dependence are at work in the Brazilian labor market, which cannot be captured by a simple static model.

Finally, in the Spanish case, the persistence of unemployment observed in the dynamic model, combined with the strong sensitivity of the labor market to economic cycles, suggests that institutional and structural factors play a decisive role in unemployment dynamics. Hysteresis effects (Blanchard and Summers, 1986), labor market duality (Bentolila and Bertola, 1990), the fiscal consolidation policies implemented after the 2008 crisis, and the sectoral transformations linked to the collapse of the construction sector all constitute complementary explanatory factors that the Okun model alone cannot capture.

Ultimately, the answer to the central research question of this study is as follows : economic growth is an important and empirically validated factor in reducing unemployment, but it is not sufficient to fully explain labor market dynamics. To fully understand the evolution of unemployment, it is necessary to combine the econometric analysis of Okun's law with a structural, institutional, and sectoral reading of each national economy. It is from this perspective that the conclusion of this study formulates economic policy recommendations and research avenues for further analysis.

4.7 Limitations of the Study

This study presents several limitations that should be explicitly mentioned in order to interpret the results with the required caution.

First, the basic model relies on a single explanatory variable, namely annual real GDP growth. Although this variable is central to Okun's law and constitutes the natural starting point for any analysis of this relationship, it does not capture the full complexity of the labor market. Other macroeconomic variables—such as inflation, the real interest rate, foreign direct investment, public consumption, or the exchange rate—could improve the explanatory capacity of the model and make it possible to test whether unemployment is better explained by a set of factors rather than by economic growth alone.

Second, the analysis is based on annual data, which limits the precision with which the adjustments between growth and unemployment can be observed. Quarterly data would have made it possible to observe the short-term dynamics of this relationship more precisely, particularly during periods of economic shock such as the 2008 financial crisis or the COVID-19 pandemic in 2020. The latter introduced extreme values into the series, particularly for Brazil, as reflected by the very high value of the Jarque-Bera test, which calls into question the normality of the Brazilian residuals. The introduction of a dummy variable for the year 2020 could have partially corrected this effect and improved the statistical properties of the model.

Third, the estimated models make it possible to identify a statistical relationship between growth and unemployment, but they do not, by themselves, prove the existence of direct causality. The question of causality between these two variables would deserve to be addressed using Granger causality tests or more sophisticated identification approaches, such as vector autoregressive models (VAR) or instrumental-variable estimations.

Fourth, the comparison between the three countries should be interpreted with caution, since Morocco, Brazil, and Spain have very different economic, institutional, and social structures. These differences profoundly influence the way in which economic growth is transmitted to the labor market, and they cannot all be captured by a simple econometric model. Finally, the availability and quality of World Bank data may vary across countries and periods, particularly for the most recent years, which may introduce imprecision into the estimations.

Finally, it should be recalled that the results obtained highlight a statistical relationship between economic growth and the change in unemployment, but do not, by themselves, establish direct causality. Okun's law makes it possible to identify a robust empirical association, but causal analysis would require complementary methods, such as Granger causality tests, VAR models, or more advanced identification strategies.

4.8 Summary of the Discussion

The comparative discussion shows that Okun's law is empirically verified in the three countries studied, but that its intensity varies significantly across national contexts. This difference is explained by productive structures, labor market institutions, the weight of informality, and each economy's capacity to transform growth into formal jobs. Thus, the results confirm that economic growth plays an important role in the evolution of unemployment, but that it is not sufficient on its own to explain the full dynamics of the labor market.

5 Conclusion

This article aimed to empirically analyze the relationship between economic growth and unemployment through a comparative study of Okun's law applied to Morocco, Brazil, and Spain. Using annual data from the World Bank covering the period 1991–2024, the study adopted an econometric approach based on ordinary least squares, ADF stationarity tests, HAC robust corrections, and dynamic models with lagged variables.

The results obtained broadly confirm the empirical validity of Okun's law in the three countries studied. In each case, the coefficient associated with real GDP growth is negative and statistically significant. This means that an increase in economic growth is associated with a decrease in the annual change in the unemployment rate. However, the intensity of this relationship varies substantially across countries. The Okun coefficient is estimated at approximately -0.10 for Morocco, -0.29 for Brazil, and -0.45 for Spain.

This difference shows that the reaction of unemployment to growth depends on the economic structures specific to each country. Spain displays the strongest relationship, reflecting the high sensitivity of its labor market to economic cycles. Brazil occupies an intermediate position, while Morocco shows a weaker relationship, probably due to the weight of informality, the sectoral structure of the economy, and the mismatch between education and employment.

The R^2 values, ranging between 0.39 and 0.42 in the basic models, nevertheless show that economic growth explains only part of the variations in unemployment. Okun's law therefore makes it possible to identify an important relationship, but it is not sufficient to explain the entire dynamics of the labor market. Other macroeconomic and structural factors must be taken into account, such as informality, investment, productivity, public policies, and the institutional functioning of the labor market.

It is also important to recall certain methodological limitations. In the case of Spain, the variable `delta_unemployment` presents borderline stationarity at the strict 5% level, which calls for a cautious interpretation of the Spanish results. In the case of Brazil, the Jarque-Bera test indicates non-normality of the residuals, probably linked to the presence of major macroeconomic shocks over the period studied. These limitations do not call into question the general relationship observed, but they confirm the need to interpret the econometric results as statistical indications rather than as a complete explanation of unemployment.

These results have important implications for economic policy, particularly in the Moroccan case. A policy supporting growth is necessary, but it must be accompanied by structural reforms capable of making this growth more inclusive and more conducive to the creation of formal jobs. This requires, in particular, a better match between training and labor market needs, a reduction in informality, and the development of labor-intensive sectors.

Finally, this study could be extended by integrating additional macroeconomic variables, using quarterly data, analyzing other countries in the MENA region or Latin America, and conducting a comparison by sub-periods in order to assess the evolution of the Okun relationship before and after major recent economic crises.

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